

1) This encryption scheme does not have CCA security because an attacker can use the chosen-ciphertext attack to modify the ciphertext in a way that will be accepted as a valid message by the decryption algorithm, without knowing the encryption key.

1. The attacker intercepts a ciphertext $C = E(m || H(m))$ where m is some unknown plaintext.
2. The attacker flips one arbitrary bit in the ciphertext to obtain a modified ciphertext C' .
3. The attacker submits C' to the decryption oracle.
4. If the modified ciphertext C' happens to decrypt to a valid message, i.e., if the last n bits are the hash of the rest of the plaintext, then the decryption oracle will return the decrypted message m' .
5. Otherwise, if the modified ciphertext C' does not decrypt to a valid message, i.e., if the last n bits are not the hash of the rest of the plaintext, then the decryption oracle will return an error.
6. The attacker repeats steps 2-5 with different modifications of C until the decryption oracle returns a valid message.
7. Once the attacker obtains a valid message m' , the attacker can compute $H(m')$ and construct a new ciphertext $C'' = E(m' || H(m'))$.
8. The attacker submits C'' to the encryption oracle, which will encrypt it under CTR mode to obtain a new ciphertext C''' .
9. The attacker replaces C' with C''' in the original intercepted ciphertext C , obtaining a new ciphertext C'''' .
10. The attacker can now send C'''' as a valid ciphertext to the decryption oracle, which will decrypt it to m' .

This attack shows that the scheme is not CCA secure.

2) To show that the CBC-MAC construction applied to a PRP is not collision-resistant, we will provide an efficient algorithm to find collisions in the given salted hash function H .

Let (m_1, m_2, m_3) and (m_1', m_2', m_3') be two distinct message triplets such that $H(k, m_1 || m_2 || m_3) = H(k, m_1' || m_2' || m_3')$. We need to find such message triplets to prove that the given construction is not collision-resistant.

We know that $c_1 = F(k, m_1)$ and $c_1' = F(k, m_1')$. Since F is a secure PRP, it is computationally infeasible to find m_1' given c_1 and c_1' (or vice versa).

However, we can observe that $c_2 = F(k, m_2 \oplus c_1)$ and $c_2' = F(k, m_2' \oplus c_1')$ are related as follows:

$$\begin{aligned}
 c_2 \oplus c_2' &= F(k, m_2 \oplus c_1) \oplus F(k, m_2' \oplus c_1') \\
 &= F(k, m_2 \oplus c_1 \oplus m_2' \oplus c_1') \\
 &= F(k, (m_2 \oplus m_2') \oplus (c_1 \oplus c_1'))
 \end{aligned}$$

Note that $(m_2 \oplus m_2', c_1 \oplus c_1')$ is a valid input to the PRP F . Therefore, we can compute c_2 and c_2' using the following steps:

Compute $c_1 = F(k, m_1)$ and $c_1' = F(k, m_1')$

Compute $c_2 = F(k, m_2 \oplus c_1)$ and $c_2' = F(k, m_2' \oplus c_1')$

Compute $d = c_2 \oplus c_2'$

Choose a random bit string r of the same length as c_1 and c_1'

Compute $c_1'' = d \oplus c_1 \oplus r$ and $c_1''' = d \oplus c_1' \oplus r$

Compute $m_1'' = F^{-1}(k, c_1'')$ and $m_1''' = F^{-1}(k, c_1''')$

Output the message triplets $(m_1, m_2, m_3) = (m_1'', m_2 \oplus c_1'', m_3 \oplus c_2'')$ and $(m_1', m_2', m_3') = (m_1''', m_2' \oplus c_1''', m_3' \oplus c_2''')$

It is easy to see that $H(k, m_1'' || m_2 \oplus c_1'' || m_3 \oplus c_2'') = H(k, m_1''' || m_2' \oplus c_1''' || m_3' \oplus c_2''')$. Therefore, we have found a collision in the salted hash function H with non-negligible probability.